1.21 Inequalities: Identifying Regions

Chapter 3 · Original slide 30 · Inequality Regions

y

x

On graph paper, shade the region that satisfies the inequalities:

$2y + x < 14$
$y \geq x^2 - 3x - 4$

You did this at GCSE, the only difference here being that the graphs involved might not be straight lines.

Step 2:

An inequality involving x and y represents a 2D region in space.

Identify the correct side of each line each inequality represents.

Click to sketch >

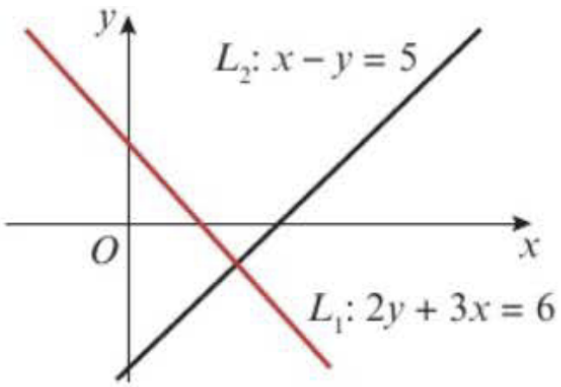
Tip: To quickly sketch $2y + x = 14$, consider what happens when x is 0 and when y is 0.

Tip: Make sure y is on the side where it is positive.

If y is on the smaller side, you're below the line.

If y is on the greater side, you're above the line.

Chapter 3 · Original slide 31 · Exercise 3F/3G – Y1 Page 53, 55



1 L_1 has equation $2y + 3x = 6$. L_2 has the equation $x - y = 5$. The diagram shows a sketch of L_1, L_2.

a Find the coordinates of P, the point of intersection.

b Hence write down the solution to the inequality $2y + 3x > x - y$.

1.21 Inequalities: Identifying Regions

Chapter 3 · Original slide 32 · Exercise 3F/3G – Y1 Page 53, 55

3 Find the set of values of x for which the curve with equation $y = \mathrm{f}(x)$ is below the line with equation $y = \mathrm{g}(x)$.

a $\mathrm{f}(x) = 3x^2 - 2x - 1$
$\mathrm{g}(x) = x + 5$

b $\mathrm{f}(x) = 2x^2 - 4x + 1$
$\mathrm{g}(x) = 3x - 2$

c $\mathrm{f}(x) = 5x - 2x^2 - 4$
$\mathrm{g}(x) = -2x - 1$